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SYSTEMATIC REVIEW

Machine-learning Algorithms for Ischemic Heart Disease Prediction: A Systematic Review

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Abstract: ***Purpose*:** This review aims to summarize and evaluate the most accurate machine-learning algorithm used to predict ischemic heart disease.

***Methods*:** This systematic review was performed following PRISMA guidelines. A comprehensive search was carried out using multiple databases such as Science Direct, PubMed\ MEDLINE, CINAHL, and IEEE explore.

***Results*:** Thirteen articles published between 2017 to 2021 were eligible for inclusion. Three themes were extracted: the commonly used algorithm to predict ischemic heart disease, the accuracy of algorithms to predict ischemic heart disease, and the clinical outcomes to improve the quality of care. All methods have utilized supervised and unsupervised machine-learning.

***Conclusion*:** Applying machine-learning is expected to assist clinicians in interpreting patients' data and implementing optimal algorithms for their datasets. Furthermore, machine-learning can build evidence-based that supports health care providers to manage individual situations who need invasive procedures such as catheterizations.

This review is registered at PROSPERO; the registration number is: CRD42021288599.

Keywords: Big data, data mining, machine-learning algorithms, prediction, ischemic heart disease, PRISMA.

1. INTRODUCTION

The World Health Organization (WHO) estimated that in 2019 around 18 million people died due to heart diseases representing about 32% of death worldwide [1]. Acute Myocardial Infarction (MI) along with unstable and stable angina are known as Ischemic Heart disease (IHD) characterized by necrosis or complete occlusion of the main coronary artery /arteries which slows the blood supply to specific parts of heart musculature caused by plaque rupture and formation of clots known as a thrombus [2]. The American Heart Association (AHA) estimates that one person in the United States experiences an acute MI every 40 seconds, which raises the financial burden of healthcare because of the high morbidity and mortality rate [3].

Heart diseases can go undetected and show no symptoms at all until a patient has a heart attack, at which point they manifest as chest pain, heartburn, indigestion, shortness of breath, palpitations, and swellings of the feet, ankles, and belly [4]. In order to adequately act for people at risk for cardiovascular illnesses, early detection and prediction of IHD are therefore essential [5].

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The goal of Electronic Health Records (EHRs) is to improve the efficiency of health management, raise the bar of treatment, and streamline workflow processes in healthcare settings [6]. Making an accurate diagnosis of cardiac disorders depends on several contributing risk factors and laboratory findings [7]. In this enormous dataset, it is beneficial to find patterns that can be used to forecast individuals who are at high risk of developing diseases of the heart [8] using machine-learning algorithms and computers.

Machine-learning algorithms are technological applications that are used to support clinical decisions for several medical cases [4]. For instance, kidney diseases were predicted using classification techniques of machine-learning, namely, Naïve Bayes (NB) and Artificial Neural Network (ANN) [5], which produce more accurate results using the NB than ANN. Another study utilized NB and Decision Tree (DT) for optimizing liver disease classification to find promising results with the fastest computational time using the NB algorithm [6].

In addition, many models (algorithms) were used in big data analytics and machine learning to predict IHDs. Different traits for each model were defined, including strong, straightforward, accurate, particular, sensitive, and others. Therefore, the objective of this study was to assess recent articles that used machine-learning methods to identify and forecast IHD. Additionally, by analyzing the benefits and drawbacks of each model used to predict IHD, this research

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will aid medical professionals in reaching the most informed clinical judgments based on patient data.

2. MATERIALS AND METHODS

This systematic review was guided by the Preferred Reporting Items as the best-recommended methods for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [7].

2.1. Searching Methods

The search was conducted to identify the most frequent models used in machine-learning algorithms to predict IHD using Science Direct, PubMed\MEDLINE, CINAHL, and IEEE explore for articles published between 2017 and 2021. The search terms included were "ischemic heart disease", "machine-learning algorithms", "myocardial infarction", "big data", and "prediction". A combination of key terms was used. Medical Subject Heading (MeSh) and the Boolean operators were used while searching in the initial search of machine-learning and prediction of IHD concepts. Furthermore, the lists of references for the reviewed articles were also examined for other potentially appropriate articles.

2.2. Inclusion Criteria

Articles that utilize machine-learning algorithms to predict IHD if met the following criteria were included: (1) utilize or build a model(s) to predict IHD (2) use measurements such as specificity, sensitivity, precision, accuracy, and F-score to assess the precision of the model (s) used; (3) written in the English language; (4) published in the period between 2017 and 2021. While unpublished articles, grey literature, student's dissertations, thesis, book chapters, encyclopedias, and conference papers were excluded from the search.

2.3. Study Selection and Data Extraction

The title and abstract of the articles under evaluation were first independently checked by the two authors of this study. To obtain a consensus, the two authors discussed any points of contention. Duplicate entries and items that caused the dispute were removed. Then, the two authors extracted information that was pertinent to the fundamental characteristics, such as the names of the authors, publication dates, models employed in the study, and prediction results.

2.4. Synthesis of Results

The reviewers finished the data synthesis and analysis; any discrepancies between them were settled by consensus. Tables and a synthesis of the evidence, highlighting the consistency of key findings across research, were created. Using the Critical Appraisals Skills Program Checklist, the two reviewers assessed the likelihood of bias in each study [8].

2.5. Study Selection

As shown in Fig. (**1**), the screened articles that resulted in the preliminary search were 237. Thirteen studies were retained after filtering articles based on the eligibility criteria. Five of the reviewed articles (38.5%) were published in 2021.

3. RESULTS OF INDIVIDUAL STUDIES

Table **1** presents a matrix for the reviewed studies. All categories were identified based on the thematic synthesis of the available studies.

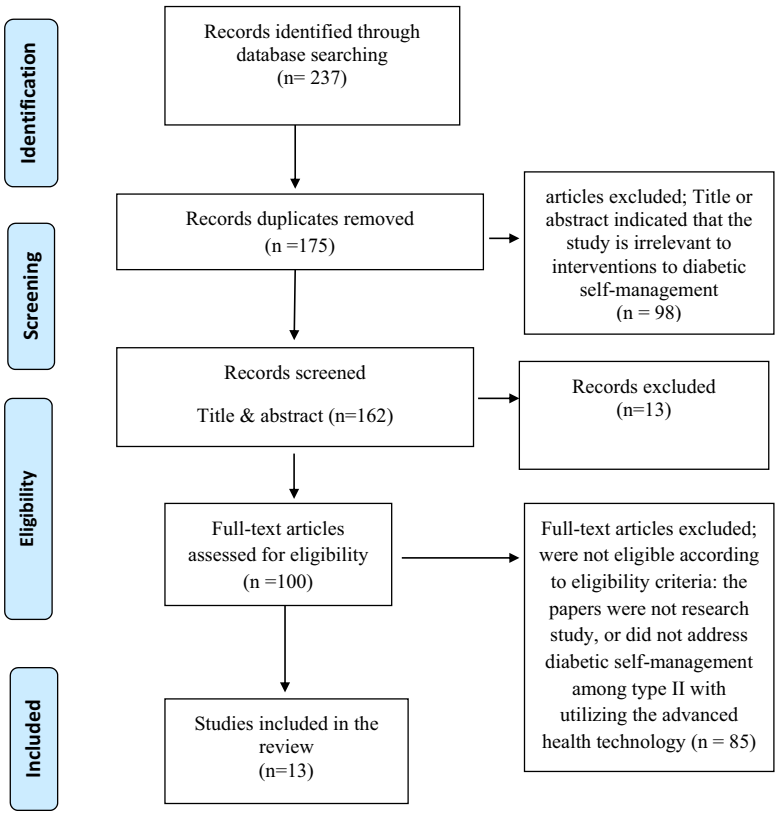

Fig. (1). PRISMA flowchart.

Table 1. Review matrix.

Author, Year, and Country	Title	Objective	Study Design	Setting and Sample	Findings	Recommendations	Additional Comments
Hend Mansoor, Islam Y. Elgendy, Richard Segal, Anthony A. Bavry, Jiang Bian. 2017. USA	Risk prediction model for in-hospital mortality in women with ST-elevation myocardial infarction: A machine-learning approach	Develop and validate prediction models for all-cause in-hospital mortality in women admitted with STEMI using logistic regression and random forest, and to compare the performance and validity of the different models	Validation cohort and prediction models using LR and RF.	Data were collected from National Inpatient Sample (NIS) and the sample included 9637 women.	Three main models were developed included: LR (with 11 variables), full RF (with 32 variables) and reduced RF(with 11 variables). In the internal validation cohort, the C-index was 0.84, 0.81, and 0.80 for the multivariate logistic regression, full, and reduced random forest models, respectively.	RF was comparable to LR in predicting in-hospital mortality in women with STEMI, and can be a useful and accurate tool in clinical practice	-
Yaowang Lin, Shaohong Dong, William W. Chu, Xianmin Kwauk, Xudong Liu. 2018. China	Applications of Machine-learning Algorithms in Predicting Coronary Artery Disease and Myocardial Infarction	developing effective methodologies and tools in CAD early diagnosis and predictive models.	Building predictive model.	4,049 complete medical records were collected from Cardiology Department, the Hospital, the People's Republic of China. Among them 2,678 patients were diagnosed as CAD positive, and 1,371 patients as CAD negative.	XGBoost-based ML model demonstrated the AUC of 99.8% for CAD and 90.8% for MI, which indicate that this model can be used as effective, time saving, reliable alternative to convention diagnostic measures.	AI machine-learning models can reliably predict CAD risks with high accuracy.	Large population of people may potentially benefit from the ML models, by seeking early diagnosis and treatment, and consequently their morbidity and mortality from CAD and MI can be significantly decreased.
Sushmita Roy Tithi, Afifa A Aktar, Fahimul A leem, A mitabha Chakrabarty. 2019.	ECG data analysis and heart diseases prediction using machine-learning algorithms	Figured out which algorithm gives the best result in predicting the diseases.	Comparative design.	UCI Machine-learning Repository database were used recruiting 452 instances with 249 attributes.	For all cardiac disease it was found that NB has the highest score of more than .9. For MI it was found that DT has a score of 96%.	-	-
M. Raihan, Parichay Kumar Mandal, et al.,. 2019. Bangladesh.	Risk Prediction of Ischemic Heart Disease Using Artificial Neural Network	To predict IHD more accurately using artificial neural network and ML techniques, more specifically predicting the heart attack.	Predictive design	The data of 835 instances with 13 features having ECG report and diagnosis result are extracted from Fortis Escorts Heart Institute, Khulna, Bangladesh.	ANN was applied to find that the risk prediction is determined around 84.47%. The precision, sensitivity, specificity, and f1-score, for the ANN, are 79.97, 82.69, 72.63, and 85.17 respectively.	The significant contingency of tachycardia is anticipated using the ANN.	Intelligent algorithm tool can contribute the enhancement of the accuracy of disease treatment and playing an appearance as a prevention in the interim of tachycardia.

(Table 1). Contd…

Author, Year, and Country	Title	Objective	Study Design	Setting and Sample	Findings	Recommendations	Additional Comments
Martin P. Than, John W. Pickering, et al,, Mills. 2019. New Zealand	Machine-learning to Predict the Likelihood of Acute Myocardial Infarction	To combine variables of troponin concentration by age, sex, and time between samples in patients with suspected MI through machine-learning to improve the assessment of risk for individual patients.	A machine-learning algorithm (myocardial-ischemic-injury-index [MI[3]]) incorporating age, sex, and paired high-sensitivity cardiac troponin I concentrations using gradient boost.	The sample was trained on 3,013 patients and tested on 7,998 patients with suspected MI.	Gradient boosting used to compute a value (0-100) reflecting an individual's likelihood of a diagnosis of type 1 MI and estimates the sensitivity, NPV, specificity and PPV for that individual. The finding represents one of the first effective demonstrations of how ML could be used to guide clinical decision making in patients with suspected acute coronary syndrome.	The MI (injury-infarction-index) clinical decision support tool incorporates simple and objective variables including age, sex and serial cardiac troponin I concentrations measured using a high-sensitivity assay to rapidly estimate risk of MI. It can be used to individualize the risk assessment of patients with suspected MI or to categorize patients into low- or high-risk groups.	Using MLA, MI provides an individualized and objective assessment of the likelihood of MI, which can be used to identify low-risk and high-risk patients who may benefit from earlier clinical decisions.
Yifan Zhao a, Jing Xiong, et al., 2020. China	Early detection of ST-segment elevated myocardial infarction by artificial intelligence with 12-lead electrocardiogram.	Building a diagnostic algorithm for detecting STEMI based on raw 12-lead ECG in consecutive STEMI patients and controls	Retrospective study.	Data collected from inpatients in the cardiology department of Shanghai Tenth People's Hospital and Changhai Hospital from 2008 to 2018. 667 STEMI samples validated by coronary angiography compared with 7571 control ECG.	Res-Net was the chosen model to fit the chosen data.	The algorithm that was used achieved AUC of 0.9954 (95% CI, 0.9885 to 1) with sensitivity specificity, accuracy, precision and F1 scores of 96.75%, 99.20%, 99.01%, 90.86% and 0.9372 respectively. In a comparative test with cardiologists, the algorithm had an AUC of 0.9740 (95% CI, 0.9419 to 1), and its sensitivity, specificity, accuracy, precision, and F1 score were 90%, 98% and 94%, 97.82% and 0.9375 respectively, while the medical doctors had sensitivity, specificity, accuracy, precision and F1 score of 71.73%, 89.33%, 80.53%, 87.05% and 0.8817 respectively.	

(Table 1). Contd…

Author, Year, and Country	Title	Objective	Study Design	Setting and Sample	Findings	Recommendations	Additional Comments
Syed Waseem Abbas Sherazi Yu Jun Jeong Moon Hyun Jae Jang-Whan Bae Jong Yun Lee. 2020. Korea	A machine-learning–based 1-year mortality prediction model after hospital discharge for clinical patients with acute coronary syndrome	Propose a ML–based 1-year mortality prediction model after discharge in clinical patients with ACS.	Building a predictive model and compare ML algorithms with GRACR risk scores.	Korea AMI Registry data set, a cardiovascular disease database registered in 52 hospitals in Korea for 1 November 2005–30 January 2008 and selected 10,813 subjects with 1-year follow-up.	The study apply 4 ML algorithms including GBM, GLM, RF, and DNN which perform that GBM is the higher performance in prediction mortality of ACS.		The performance in machine-learning–based mortality prediction model were superior to GRACE risk scores.
Johannes T Neumann, Nils A So rensen, et al., . 2020. USA	Application of a machine-learning-driven, multi-biomarker panel for prediction of incident cardiovascular events in patients with suspected myocardial infarction	Validate a machine learning-driven, multibiomarker panel for prediction of incident major adverse cardiovascular events in patient with suspected MI.	Cohort design using prognostic panel	A sample of 748 patients recruited from University Hospital Hamburg-Eppendorf.	Logistic regression using prognostic score was used to predict patient at high risk to develop MI.	The study validate the high accuracy of a multiple biomarker panel to predict incident cardiovascular events in patients with suspected myocardial infarction.	-
Divneet Mandair, Premanand Tiwari, Steven Simon, Kathryn L. Colborn and Michael A. Rosenberg. 2020. USA	Prediction of incident myocardial infarction using machine-learning applied to harmonized electronic health record data	Test the efficacy of ML methods and compare ML with traditional statistical approaches.	Building predictive models	2 million subjects and 52,000 Features were used to build predictive models for MI. The study took place in UCHealth hospital system that includes 3 large regional centers (North, Central, South) over the front range of Colorado.	DNN with random under sampling had the most accurate classification with an F score of .092 and AUC of .835.	-	The study use the following prediction model: NB, LR, R, shallow and deep NN, GBM.

(Table 1). Contd…

Author, Year, and Country	Title	Objective	Study Design	Setting and Sample	Findings	Recommendations	Additional Comments
Yao Wang, Kangjun Zhu, Ya Li, Qingbo Lv, Guosheng Fu, Wenbin Zhang. 2020. China	A machine-learning-based approach for the prediction of periprocedural myocardial infarction by using routine data	Evaluate whether ML can develop a robust prediction model for PMI, and determine which class of ML algorithm has the highest predictive potential.	To build a predictive model	The dataset was based on the inpatients who were admitted in department of cardiology of Sir Run Shaw hospital in China from the period of 2007 to 2019. The sample included total of 10,886 cardiac inpatients.	ANN and RF showed the most accurate performance in PMI3 and PMI5.	ML methods may provide accurate prediction of PMI in CAD patients	ML could be used as a precise model in the preventive treatment of PMI.
Han Cheol Lee, Jeong Cheon Choe, Jin Hee Ahn, Hye Won Lee, Jun-Hyok Oh, Jung Hyun Choi Kwang Soo Cha, Taek Jong Hong, and Myung Ho Jeong. 2020 Korea	Prediction of 1-Year Mortality from Acute Myocardial Infarction Using Machine-learning	Develop and validate a 1-year mortality risk model using machine-learning, and compared the performance to an existing risk model.	Predictive design	Korea Acute Myocardial Infarction Registry (KAMIR) dataset was selected to recruit 22,182 AMI patients.	Comparison between KAMAIR score model and ML models was done to find that ML models had higher specificity and sensitivity predictive outcome to define patient at high risk mortality rate AMI.	ML models could support risk stratification and management in post discharge AMI patients.	ML that used were DT, ensembles, LR, deepnets algorithms.
Zhixun Bai , Jing Lu, Ting Li, Yi Ma, Zhijiang Liu, Ranzun Zhao, Zhenglong Wang, and Bei Shi. 2021. China	Clinical Feature-Based Machine-learning Model for 1-Year Mortality Risk Prediction of ST-Segment Elevation Myocardial Infarction in Patients with Hyperuricemia: A Retrospective Study	Evaluate the performance of different ML models for the prediction of 1-year mortality in STEMI patients with hyperuricemia.	Retrospective design	656 patients were enrolled in the study in Zunyi Medical University between January 2016 and January 2020.	Five machine-learning algorithms (logistic regression, KNN, RF, XGBoost, and CatBoost) were developed to predict the 1-year mortality rate with all available features to find that CatBoost prediction performance is higher than traditional GRACE risk scores.	The predictive ability of machine-learning methods is significantly higher than that of the traditional statistical scoring model.	CatBoost has an accuracy of 0.89, AUC 0.87).
Firdaus Aziz, Sorayya MalekID, et al., 2021. Japan	Short- and long-term mortality prediction after an acute ST-elevation myocardial infarction (STEMI) in Asians: A machine-learning approach	Apply ML for the prediction and identification of factors associated with short and long-term mortality in Asian STEMI patients and compare with a conventional risk score.	Retrospective study.	The National Cardiovascular Disease Database for Malaysia registry, used for in-hospital (6299 pt.'s) 30-days (3130 pt.'s), and 1-year (2939 pt.'s)) model development. 50 variables were considered.	SVMvarImp-SBE-SVM for in-hospital, 30 days and 1-year mortality prediction had better performance compared to RF, LR and TIMI scoring as well.	The study perform the capability of ML algorithms application for feature selection and prediction of in-hospital, 30 days and 1-year population-specific mortality in STEMI patients.	ML models that are population-specific demonstrate better performance compared to TIMI score.

(Table 1). Contd…

Author, Year, and Country	Title	Objective	Study Design	Setting and Sample	Findings	Recommendations	Additional Comments
Woojoo Lee, Joongyub Lee, Seoung-Il Woo, Seong Huan Choi, Jang-Whan Bae, Seungpil Jung, Myung Ho Jeong & Won Kyung Lee. 2021. Korea.	Machine-learning enhances the performance of short and long-term mortality prediction model In non-ST-segment elevation MI.	Developed ML-based models (LR with regularization, RF, SVM, and XG boosting) and compared their performance in predicting the short- and long-term mortality of patients with AMI with those of TMs with comparable predictors.	Retrospective cohort study	15,247 participants were enrolled from Korean Registry of Acute Myocardial Infarction for Regional Cardio-cerebrovascular Centers (KRAMI-RCC) registry.	ML algorithms, including RF, SMV, XGBoost, Lasso, Ridge regression, and Elastic net, were applied to develop a mortality prediction model. It was found that XGboost is the best prediction model with higher performance of NSTEMI.	The performance of the predictive model could be improved, particularly for long-term mortality in NSTEMI, from the ML algorithm rather than using more clinical predictors.	-
Sana Al Azwari. 2021. Saudi Arabia	Predicting Myocardial Rupture after Acute Myocardial Infarction in Hospitalized Patients using Machine-learning	Employing the Random Forest model from the MLA to predict myocardial rupture after acute MI in hospitalized patients.	Prediction model (RF)	Database for this specimen was collected in the Krasnoyarsk Interdistrict Clinical Hospital No20, named after I. S. Berzon. (Russia) in 1992-1995. The data contains 1700 records (patients), 111 input features, 12 complications.	RF regression was take the age and sex at the most attributing predictors of developing myocardial rupture after MI in hospitalized patients.	The advantage of AI algorithms is that they can save millions of lives. This innovative method aims to integrate automation in machine-learning algorithms to achieve a highly educated guess based on the data provided.	The prediction of Myocardial Rupture in patients with Myocardial Infarction could bring rapid awareness and educate people about the necessary changes in their lifestyles along with releasing the burden of hefty expenditures from the national exchequer.
Jia Zhao, Pengyu Zhao, Chunjie Li Yonghong Hou. 2021. China	Optimized Machine-learning Models to Predict In-Hospital Mortality for Patients with ST-Segment Elevation Myocardial Infarction	To propose a dataset that can represent the real-world STEMI patients, apply RUS to optimize prediction models, and develop models that can predict high-risk patients at the time of initial evaluation and treatment.	predictive model including LR, SVM, RF, DT.	The study was conducted using a hospital-based dataset recruited consecutive sample (5708) STEMI patients in Tianjin Chest Hospital from January 2015 to April 2020	RUS was applied to optimize the models that used in the study.	RUS using alleviate the effect of data imbalanced, which improved the performance of models. For models trained with the full set, the SVM achieved the best performance. While for models trained with the simplified set, the SVM achieved better outcomes comparable to the models trained using the full set.	The study provides an effective and robust method for predicting in-hospital mortality of STEMI patients.
Ladda Ashish, Sravan Kumar V, Sahithi Yeligeti. 2021. India	Ischemic heart disease detection using support vector Machine and extreme gradient boosting method	To develop an early detection and localization system for IHD based on Magneto –Cardio Graphy which is reliable method.	Building a MLA and compare the results of detecting IHD.	Z-Alizadeh sani Heart Dataset (HD) was used along with 164 features of T-T intervals of ECG.	Comparison between prediction models were demonstrating using the accuracy and F-score (RF, SVM, XG Boost, and mix between both SVM and XG Boost).	Among the 3 entity classifier processes that is mentioned, the mixed model of SVM-XG-Boost proves best presentation with precision of (93.86%), and F1 score of (91.86%).	-

3.1. Identification of Commonly used Algorithm (s)\ Model (s) to Predict IHD

To determine the most accurate model for the prediction of cardiovascular disorders like IHD and locate new risk variables that could cause IHD, many machine-learning algorithms were used. For example, ANN is one of the machine-learning models that could be used to achieve helpful results. By implementing ANN for IHD patients, the risk prediction was around 84.47% [9]. Another study that assessed the risk for both MI and Coronary Artery Disease (CAD) utilizing the XGBoost found that both cases achieved high accuracy of 99.8% area under the curve for CAD and 90.8% for MI [10]. Six algorithms named: LR, DT, KNN, NB, ANN, and Support Vector Machine (SVM) to distinguish normal and abnormal electrocardiograms by analyzing them. It was found that CAD is best detected by the NB algorithm while DT is the best method to detect MI [11].

Extreme Gradient Boosting (XGBoost) was found to be the best performing model with 0.912 of the area found under the curve, outperforming the traditional model that used regression-based risk prediction to improve the prediction model of short and long-term mortality in Non-ST-Segment Elevation MI (NSTEMI). [12]. On the other hand, a traditional method of LR and other machine-learning algorithms of Random Forest (RF), XGBoost, and shallow and deep neural networks were used to predict the incidence of MI to find that the traditional method of LR produced benefit results in predicting the outcome of 6-month MI incidence. However, the calibration of all applied models was poor because there was overfitting from the low frequency of the event of MI [13].

A study by Bai and colleagues [14] on various machine-learning models utilizing LR, KNN, CatBoost, RF, and XGBoost aimed at evaluating their performance to predict one-year mortality prediction among ST-Segment Elevation MI (STEMI) complicated by hyperuricemia. Overall, it was found that the CatBoost model achieved the highest accuracy of 89%, which can predict the mortality for patients with STEMI who were complicated by hyperuricemia.

Both SVM and XGBoost, which were classified using the RF to train and test the dataset to find IHD instances, produced better prediction results. The utilized data set extracted 146 features of T-wave which were segmented from averaged Magneto –Cardio Graph (MCG) to find that accuracy of detection of IHD cases reached 93.08%, with the highest F-score of .915 indicating N2Genetic-SVM which is a combination of XGBoost and SVM producing high-performance ability to predict IHD [15].

3.2. Accuracy of Algorithms in Predicting IHD

The core of data science is machine learning, where various models (algorithms) are developed for use in the medical field [16]. Machine learning has two approaches to label data known as supervised and unsupervised machine-learning models [17], in which the supervised machine learning models are used to learn, interpret and predict the outcomes from the training dataset which requires upfront human intervention to label the given data appropriately while in unsupervised learning work on their own to predict and discover the structure of unlabeled data [18]. However, performance evaluation of utilized models was judged using the accuracy, also known as sensitivity, specificity, F1-score, and Area Under the Curve (AUC) [9]. Additionally, calibration and discrimantion were used to determine how closely the risk prediction models' predictions corresponded to the actual results of the provided data [19].

Generally, accuracy is known as the quantity of total used datasets which are correctly predicted out of the total cases [accuracy= (true positive+ true negative)/total], sensitivity is the number of positive dataset cases of the actual positive cases [sensitivity = true positives/(true positives + false negatives)], specificity is the quantity of actual negative cases which were predicted as the negative [True Negative/(True Negative + False Positive)], precision is the proportion of positive outcome that are actually positive predicted samples [precision = True positive/true positive+false positive], the F1 score is a combined harmonic average or the average of reciprocals that merges both precision and recall and it is calculated as [F1 = 2 × (precision × recall)/(precision + recall)], and AUC is an estimation of the possibility that a model categorizes a randomly chosen positive instance higher than at a randomly chosen negative cases, and it can be used to compare the performance of models visually [4].

A number of methods that use machine learning have a substantial impact on disease diagnosis and prediction [5, 20]. Furthermore, the application of machine-learning to predict cardiovascular diseases is widely used and it depends on the availability of an appropriate dataset, but there is no single method to verify the accuracy of the utilized machine-learning algorithm. Hence, a comparative analysis of IHD prediction was conducted based on the performance measures (Table **2**).

3.3. Supervised Machine-learning

A Decision Tree (DT) is a graph that visualizes choices and their results in the form of a tree. The tree is composed of nodes and branches, in which the node shows the attributes in a group while the branch reveals the value in which the node can take [18].

Extreme Gradient Boosting (XGBoost) is a scalable extreme gradient decision tree that provides a parallel tree boosting used for regression, classification, and ranking of ensemble datasets [21]. In a study that recruited 4049 medical records from a cardiology center, XGBoost was used to diagnose a model to predict patients prone to develop CAD and MI. The collected data were divided into training and testing datasets performing (3,239) and (810) cases, respectively. Based on a prediction value for both CAD (97.7%) and MI (80.1%) there was a significant difference in the accuracy of predicting the risk of MI since it develops in a different time scale and is considered a complication of CAD. However, it was concluded that the improvement in predicting accurate results with a large clinical dataset [10].

Similarly, the XGBoost model was developed utilizing variables of age, sex, and troponin time concentration at the time of MI presentation to appraise the probability of the diagnosis of MI. The study enrolled 11,011 participants from nine countries whereas a total of 3,013 were used to train the

Table 2. Performance metrics of machine-learning algorithms to predict ischemic heart disease.

Study Title	Model(s)		Performance Measures				
		Accuracy %	Sensitivity %	Specificity %	Precision %	F- Score %	AUC
Risk Prediction of IHD Using ANN	ANN	84.47	82.69	72.63	79.97	85.17	-
Applications of MLA in Predicting CAD and MI	XGBoost CAD	97.7	99.1	94.9	97.4	98.2	99.8
	XGBoost MI	80.1	90.8	71.1	72.6	80.7	90.8
Early detection of STEMI by AI with 12-lead ECG	ResNet	99.01	96.8	99.2	90.9	93.71	99.5
Machine-learning enhances the performance of short and long-term mortality prediction model in NSTEMI	LR	87.3	72.6	88.1	-	38.8	89.0
	RF	81.7	82.3	81.7	-	33.3	91.0
	SVM	79.7	67.7	80.4	-	27.1	81.9
	XGBoost	84.5	83.9	84.5	-	37.6	0.912
Prediction of incident MI using ML applied to harmonized electronic health record data	NB	-	-	-	-	60.0	0.73
	RF	-	-	-	-	84.0	0.765
	DNN	-	-	-	-	92.0	0.835
	LR	-	-	-	-	84.0	0.829
Clinical Feature-Based Machine-learning Model for 1-Year Mortality Risk Prediction of STEMI in Patients with Hyperuricemia: A Retrospective Study	CatBoost	96.0	98.0	-	95.0	97.0	0.99
	RF	95.0	98.0	-	94.0	96.0	0.99
	XGBoost	94.0	98.0	-	92.0	95.0	0.98
	LR	91.0	92.0	-	92.0	92.0	0.95
	KNN	92.0	98.0	-	.98	93.0	0.96
A machine-learning-based approach for the prediction of periprocedural MI by using routine data	SVM	69.0	73.0	-	-	69.0	0.76
	LR	71.0	67.0	-	-	69.0	0.76
	RF	71.0	70.0	-	-	70.0	0.77
	ANN	72.0	72.0	-	-	72.0	0.77
Machine-learning to Predict the Likelihood of Acute MI	XGBoost	-	89.6	89.3	-	-	0.963
Short- and long-term mortality prediction after an acute STEMI Asians: A machine learning approach	RF	93.5	34.7	96.8	-	-	0.860
	LR	84.2	71.3	85.0	-	-	0.850
	SVM	87.7	61.4	89.2	-	-	0.870
A machine-learning–based 1-year mortality prediction model after hospital discharge for clinical patients with ACS	GBM	**94.7**	**97.7**	-	96.7	**97.2**	.898
	GLM	92.6	94.9	-	97.2	96.0	.873
	RF	93.5	95.4	-	97.6	96.5	.883
	DNN	91.1	92.7	-	97.7	95.1	.898

(Table 2). Contd…

Study Title	Model(s)		Performance Measures				
		Accuracy %	Sensitivity %	Specificity %	Precision %	F- Score %	AUC
Optimized Machine-learning Models to Predict In-Hospital Mortality for Patients with STEMI	LR	73.8	89.5	73.3	-	-	.908
	SVM	85.6	84.2	85.7	-	-	.919
	DT	78.3	81.6	78.2	-	-	.819
	RF	71.7	81.6	71.4	-	-	.809
IHD detection using SVM and XGBoost method	RF	80.2	-	-	-	82.64	-
	XGBoost	89.0	-	-	-	79.86	-
	SVM	84.3	-	-	-	86.29	-
Risk prediction model for in-hospital mortality in women with STEMI: A MLA	LR	89.0	-	-	65.0		0.84
	Full RF	89.0	-	-	60.0		0.81
	Reduced RF	88.0	-	-	43.0		0.80

model while 7,998 were used as the test datasets. It was found that MI happened among 404 patients who were in the training set while it occurred among 849 patients who were enrolled in the test set. Patients with and without MI were compared in a primary calibration and discrimination study. Similar performance was estimated in the early and late presentations of serum troponin concentration with a measure of 0.963 for AUC [95% CI 0.957-0.968], the sensitivity was 89.6%, and specificity was 89.3% [22].

Another member of XGBoost is known as the CatBoost technique on DT has been utilized to predict 1-year mortality among patients with STEMI complicated with hyperuricemia. The study compares the prediction performance using five models namely, CatBoost, XGBoost, RF, LR, and KNN, enrolling 656 patients to find that CatBoost performed the highest predictive accuracy 96.0%, precision of 95.0%, and a discrimination of AUC at 99.0%. It was determined that the CatBoost model can precisely predict the mortality associated with STEMI among patients who have hyperuricemia after 1-year of development in their situation [14].

Another study involved 14,885 individuals with acute coronary syndrome (ACS), and it compared the risk of mortality among those patients to the Global Registry of Acute Coronary Events (GRACE) model using four machine-learning models: GBM, Generalized Linear Model (GLM), DNN, and RF. With an AUC of.898, the data demonstrated that GBM outperformed measures with an accuracy of 94.7 percent. Additionally, it was projected that regression mortality prediction using the GRACE model was outperformed by machine-learning-based mortality prediction [23].

In a study, IHD cases were collected from the Z-Alizadeh Sani Heart Dataset and IHD cases were detected using machine-learning classifiers. In comparison to SVM, which had an accuracy of 84.3 percent, it was discovered that XGBoost performed with the highest accuracy, at 89 percent. However, a proposed integrated model of SVM-XGBoost shows the best performance of finding for accuracy of 93.86%, and F1 score of 91.86% in prediction IHD cases [15].

Logistic Regression is used to calculate or predict the possibility of an event occurring [17]. A study was directed to develop machine-learning using RF, DT, SVM, and XGBoost and compare the performance measures of these models with the conventional models. Selection and training of the model were conducted with 70% as a training dataset after creation and validation of data and the rest of the data (30%) were used as a testing dataset. Instead of using the GRACE clinical predators, it was found that machine learning could reduce the long-term mortality of patients with NSTEMI. Additionally, it was discovered that LR, which has an accuracy of 87.3 percent, is the most accurate forecasting model among the ones that were utilized. This is in contrast to XGBoost, RF, and SVM, which have accuracy values of 84.5 percent, 81.7 percent, and 79.7 percent, respectively [12]. The GRACE as a traditional model shows excellent discrimination with an AUC of 0.91 (95% CI, 0.891–0.950). However, the performance of the selected predictive model can be improved for NSTEMI in long-term mortality rather than using the clinical prediction measures.

Multivariate LR has an excellent performance tool in clinical practice, with an accuracy of 89 percent and an AUC of 0.84, according to a study that was undertaken to enhance and confirm a prediction model for in-hospital mortality for STEMI among 9637 women using LR, full and reduced RF [24].

When faced with classification and regression problems, Support Vector Machine (SVM) is used. Each data case is created in n-dimensional space with a unique value for each variable.

A total of 12,368 were divided into 70% for model development and 30% as validation cases; three machine-learning models were used to predict and detect risk factors for short and long-term mortality among STEMI Asian patients. Supervised machine-learning of RF, LR, and SVM was used to follow up STEMI at in-hospital admission, 30-days, and one year to reveal that SVM is the best predictor model with the least number of predictors at the three identi-

fied periods with discrimination of AUC at 0.870, the accuracy of 87.7% for in-hospital patients [25].

Similar results were found in a study that intended to improve machine's learning ability to forecast in-hospital STEMI fatality rates. Based on clinical outcomes, 5708 STEMI patients were enrolled and randomly assigned to survival or death groups. The total data set was then split into training datasets (75 percent) and testing datasets (25 percent). Using Random Under-Sampling (RUS), which enhances the display of predicting models, four models were trained and optimized. With an accuracy of 85.6 percent, a sensitivity of 84.2 percent, a specificity of 85.7 percent, and a discrimination of AUC of 0.919, it was discovered that SVM performed the best among the models used [26].

3.4. Unsupervised Machine-Learning

An Artificial Neural Network (ANN) is built on the structure and function of the natural brain in which the network learns based on the input and output data. It is composed of three layers: the input levels that receive the information for processing, the output level that has the results of processing data, and the level in between, known as a hidden level [4]. In a study conducted in a heart institute utilizing 917 instances with ECG in Bangladesh, the ANN has got accuracy using real-time dataset and is considered the best performance measure to predict IHD with an accuracy reaching 84.4%, sensitivity 82.6%, specificity 72.6%, and F-score 85.1% [9]. To assess the performance of risk prediction the threshold was used to assess the corresponding sensitivity of the model to be at a rate of true positive and false positive at the ROC at 0.82.

A particular kind of ANN called a residual neural network (ResNet) is founded on a theory that has been proven by research on the cerebral cortex's pyramidal cells. In a study that enroll a dataset composed of 667 STEMI ECG and a total of 7571 control ECG based on the ResNet model to develop an auto-diagnosis model based on ECG results. The model attained a total sensitivity reaching 96.75%, specificity 99.2%, accuracy 99.0%, precision 90.86%, and F-score 0.9371. Although ResNet has been significantly used in the medical domain, it performs better results in cases of small samples and other two-dimensional images [26]. The performance of both the internal and external validation as a calibration represented using the AUC, which was 0.974 (95% CI, 0.9419 to 1), revealing that it is possible to reach high accuracy and specificity utilizing 12-lead ECG as a diagnostic test using a small amount of data.

Another type of ANN known as DNN was utilized in a large cohort study that included a systematic examination of electronic data sampled from more than two million cases with harmonized 52,000 features utilized machine-learning algorithms of RF, LR, shallow, and deep neural network, and NB. The final dataset was split into 80% as a training dataset, hyperparameter tunning .3%, and 19.7% for testing dataset to compare the models that utilized in study. It was found that DNN is superior to other machine-learning. However, DNN revealed moderate performance measures of F score (.092) and AUC of 0.835. Besides, the calibration model of the DNN shows wide discrepancy between actual and predicted probabilities of the model, while in regression models, the calibration shows the similar discrepancy between the actual and predicted values revealing good discrimination in the predicted probabilities. The study further suggests utilizing DNN because, compared to other machine-learning models using identified risk factors for MI [17], it offers a significant advantage in identifying and predicting the condition using harmonized data.

Another study was conducted on 10,886 patients with 55 attributes aiming at developing a machine-learning model to predict periprocedural MI patients. The collected data were split into a training dataset (80%, N=8709) and a validation dataset (20%, N= 2177). The study used six different categories of data, including general patient information, medical history, biochemistry results, blood results, and procedural dynamics. Four different machine-learning were used, namely RF, SVM, LR, and ANN, to find that ANN performed the best control to predict the incidence of periprocedural MI with an accuracy of 72% and discrimination of AUC at 0.77 [27].

Clinical Outcomes to Improve Quality of Care Utilizing Machine-learning for Patients with IHD.

According to the WHO, heart diseases account for 17.9 million people annually, representing around 32% of global death. Prevention strategies such as early diagnosis and seeking the appropriate treatment using machine-learning algorithms can significantly decrease morbidity and mortality from IHD [17].

Following Percutaneous Coronary Intervention (PCI) for MI patients who have a poor prognosis after the operation, a study involving 10,886 patients was conducted. Early identification can be employed as a precise technique of therapy prevention for IHD patients, according to machine learning algorithms [32]. Another study compared the performance of prediction of STEMI infarction one-year mortality for patients complicated with hyperuricemia using a traditional predicting score model and machine-learning algorithms to find out that utilizing machine-learning has a great predictive ability for sudden deaths resulting from cardiac issues during the 1-year clinical follow-up post-discharge as a primary outcome of this study [14].

Another study found that a machine-learning model improves the performance of the models currently being used for patients with acute MI, predicting both short-term and long-term mortality for NSTEMI during in-hospital, 3-month, and 12-month periods.. Hence, physicians can take benefit from applying machine-learning to identify the patient who was at high risk for admission. Consequently, machine-learning could be merged with electronic medical records as an important source of clinical decision-making and effective utilization in the clinical field [12].

To sum up, traditional tools and laboratory findings are far away from perfectionism. So, it is crucial to improve risk prediction models using a dataset that can perform the real-time world. Wide usage of traditional risk models with the high mortality rates resulting from IHD which are far from perfect. Hence, a thriving of machine-learning algorithms could prove the evidence of enhancing both short and long-term mortality and morbidity resulting from IHD.

4. DISCUSSION

4.1. Summary of Evidence

Cardiovascular health disorders like IHD, MI, and ACS may be accurately predicted using machine learning algorithms. Applying various methods, such as supervised machine learning algorithms like DT, XGBoost, SVM, and LR and unsupervised machine learning algorithms like ANN, DNN, and KNN, led to promising findings [4, 22-32]. For supervised machine learning, the dataset has a lower number of cases with clear-labeled data that can be used to train the dataset. For instance, in a study, a total number of 4049 medical records was subdivided into training and testing datasets to reach a predictive value of 97.7% for CAD and 80% for MI prediction using the XGBoost algorithm [22]. Another study recruited only 656 patients to predict 1-year mortality among STEMI to find out that XGBoost produced a high predictive accuracy of 96% [14]. For SVM, a total of 12.368 patients were applied to predict long and short mortality among STEMI to reach an accuracy of 87.7% [25]. On the other hand, unsupervised machine learning algorithms predict high-performance results using a large-scale dataset. For example, a cohort study including more than two million cases to produce an F score of 0.092 offers a substantial benefit in classifying and predicting the IHD cases [13].

These results were consistent with a meta-analysis study that demonstrated the prognostic power of machine learning, particularly XGBoost and SVM models, in cardiovascular illnesses [33]. It is noted that supervised machine learning can predict outcomes from new data of the tested dataset, while unsupervised machine-learning it could bring specific insights from a large-scale data set. Hence, the main drawback of supervised machine-learning is that it is time-consuming to train and label the given dataset to produce accurate results, while unsupervised machine-learning can produce inaccurate results which require human intervention to judge the quality of output data [17].

However, utilizing such machine-learning algorithms can improve clinicians' ability in taking the appropriate support clinical decision-making, cost reduction by decreasing the admission and re-admission caused by the morbidity of cardiovascular diseases. The performance of health care institutions requires improvement which is reflected in the quality of patient care [4].

The development of evidence-based machine learning algorithms can help doctors, nurses, and technicians handle specific conditions that call for invasive procedures like catheterizations and PCI [14]. Utilizing machine-learning algorithms efficiently assists clinicians in paying close attention to those patients who need intensive monitoring and rigorous care at the time of hospital stay and those who need frequent follow-up with high medication compliance at discharge. The main limitation of this systematic review is the lack of the availability of articles since few studies have analyzed the performance measurements such as accuracy, precision, sensitivity, specificity, F1-score, and AUC. Furthermore, this gives the review significant value as an innovative topic.

CONCLUSION

Three factors were examined in this systematic review of 13 studies that used machine learning algorithms on IHD patients: the widely used method to predict IHD, the reliability of the algorithms used to predict IHD, and clinical outcomes to enhance the quality of care. Both supervised and unsupervised machine learnings are used in every approach. By using machine learning, physicians may be able to better understand patient data and implement the best algorithms for their dataset. It is preferable to use unsupervised machine learning algorithms with a large dataset in order to produce better classification and prediction performance than supervised machine learning algorithms for small datasets with clear labels.

LIST OF ABBREVIATIONS

AHA	=	American Heart Association
IHD	=	Ischemic Heart disease
MI	=	Myocardial Infarction
WHO	=	World Health Organization

AUTHORS' CONTRIBUTIONS

S.BH. was responsible for the literature searches and the data analysis via the thematic analysis method. M.A. conducted the dual-review process, made critical revisions to the paper. Both authors verified all the processes in conducting this systematic review and supervised the study.

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SUPPLEMENTARY MATERIAL

PRISMA checklist is available as supplementary material on the publisher's website along with the published article.

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